



11506 - JWST–Roman–Euclid Purity Project: Calibrating Slitless Redshifts for Cosmology

Cycle: 5, Proposal Category: AR

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OBSERVATIONS

ABSTRACT

We propose an archival study that combines JWST NEXUS spectroscopy with the overlapping Euclid Deep Field North (EDFN) to calibrate the purity and completeness of Euclid’s wide spectroscopic sample and to deliver selection-function models directly applicable to both Euclid and the Nancy Grace Roman Space Telescope.

JWST Proposal 11506 (Created: Friday, March 13, 2026, 4:07:13PM Eastern Standard Time) - Overview

Our approach constructs a secure NEXUS redshift catalog in the EDFN footprint using complementary JWST modes (NIRCam/WFSS grism spectroscopy in the Wide tier and NIRSpec/MOS PRISM in the nested Deep tier) and cross-validates these redshifts against Euclid grism measurements. This overlap will provide redshifts for ~ 1000 $H < 24$ galaxies, enabling robust estimates of the overall contamination fraction ($f(z)$) as well as its decomposition into line- and noise-interlopers.

We will (i) quantify purity and completeness as a function of redshift and magnitude, (ii) diagnose the causes of line mis-identifications to model the selection function, and (iii) propagate these calibrations to Roman via end-to-end simulations.

The outcome of this project will be empirical priors on $f(z)$ and validated selection function for Roman and Euclid to be used in the inference of cosmological parameters.

OBSERVING DESCRIPTION